AI-Driven Smart Fashion Retail Recommendation System

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Multi-store fashion retailers in Sri Lanka make inventory, promotion, fitting and in-store service decisions with tools that do not share data. Stock is forecast and transferred manually, promotions are targeted by broad segments without checking stock, customers cannot judge fit across brands, and staff cannot see which shoppers need help. Each gap is usually handled by a separate system, so a decision made in one area does not inform the others. This report presents an integrated AI-driven Smart Fashion Retail System that addresses this common problem through four interdependent components: (1) smart inventory and stock-flow optimization, (2) personalized marketing intelligence, (3) virtual try-on and size recommendation, and (4) GPS-based customer behavior tracking with staff assistance. The components share a FastAPI service layer, a Flutter Web front end, Firebase Authentication and a single Firebase Firestore database keyed on common product, customer, store and transaction identifiers. The team followed an iterative process of requirement analysis, data collection, preprocessing, model development, system integration and evaluation. The inventory component's XGBoost demand model achieved an R^2 of 0.9497 for daily forecasting and drives explainable, risk-scored stock-transfer recommendations. The marketing component's XGBoost redemption classifier, built on Recency, Frequency and Monetary features, reached 85.5% test accuracy and only promotes products the inventory component reports as available. The virtual try-on component predicts body measurements with mean absolute errors of 1.12–1.45 cm and maps them to brand-specific sizes with 88.4% accuracy, combined with Vertex AI garment visualization and a retrieval-augmented stylist. The tracking component classifies browsing versus transiting movement with 91.2% accuracy and raises automated staff-assistance alerts. Together the components give retail managers, marketers, staff and customers one connected, explainable decision-support platform. Validation used partially simulated data, so real-world deployment and stakeholder trials are the main next steps.

Keywords: Smart retail, Inventory optimization, Personalized marketing, Virtual try-on, Customer behavior tracking

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
BMI	Body Mass Index
CLV	Customer Lifetime Value
CRM	Customer Relationship Management
GPS	Global Positioning System
JSON	JavaScript Object Notation
LLM	Large Language Model
MAE	Mean Absolute Error
ML	Machine Learning
MLOps	Machine Learning Operations
NFR	Non-Functional Requirement
PDPA	Personal Data Protection Act
POS	Point of Sale
R^2	Coefficient of Determination
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
RFM	Recency, Frequency and Monetary
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
SOS	Emergency assistance request
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Apparel is one of the most important sectors of the Sri Lankan economy, contributing a large share of industrial exports and formal employment [1], [2]. Alongside export manufacturing, a growing domestic fashion retail market operates through chains of physical stores supported by online channels. These retailers manage thousands of products across several brands, sizes and branches, and they compete on having the right stock in the right store, on promotions that reach the right customers, and on the quality of the in-store experience.

Artificial Intelligence (AI) and Machine Learning (ML) have turned apparel retail into a real-time, multichannel and data-driven industry [3], [4]. Gradient-boosted trees and random forests are now standard tools for retail demand forecasting [5], [6], [7]. Recency, Frequency and Monetary (RFM) and Customer Lifetime Value (CLV) frameworks support targeted promotion [8], [9]. Computer vision and generative models make virtual try-on possible [10], [11], and location data from mobile devices can reveal a shopper's intent from their movement [12].

In practice, however, these capabilities are usually adopted one at a time. A retailer may run a forecasting spreadsheet, a separate campaign tool, an online size chart and no in-store analytics at all. Because the systems do not share data, a promotion can push a product that is out of stock, a stock transfer ignores an upcoming campaign, and the behavior of customers inside the store never informs either. This project set out to build these capabilities as one integrated platform for a multi-store fashion retailer.

1.2 Stakeholders and user needs

The integrated system serves four primary stakeholder groups and several secondary ones. Table 1.1 summarizes their needs and the components that address them.

Table 1.1: Stakeholders, needs and responsible components

Stakeholder	Type	Principal needs	Components
Inventory and store managers	Primary	Reliable demand forecasts; early shortage warnings; justified, low-cost stock transfers	C1, C2
Marketing team	Primary	Know which customers are likely to redeem an offer; time campaigns; produce promotional assets quickly; avoid promoting unavailable stock	C2, C1
Customers (shoppers)	Primary	Correct size across brands; see how a garment looks before buying; relevant offers; quick help in store	C3, C2, C4
Store staff	Primary	Know which customers need assistance and where they are in the store	C4
Retail management	Secondary	Consolidated, explainable insight to support decisions across branches	All
Logistics personnel	Secondary	Clear, approved transfer instructions between stores	C1
System administrators	Secondary	One maintainable platform with managed access and data	All

Across these groups, three needs recur: decisions should be **predictive** rather than reactive; they should be **explainable**, so that managers can trust and override them; and they should be **consistent across functions**, so that marketing, inventory and in-store service act on the same data.

1.3 Alignment with the Sustainable Development Goals

The project aligns with three of the United Nations Sustainable Development Goals (SDGs) [13]:

- **SDG 12 – Responsible consumption and production.** Accurate demand forecasting and inter-store transfers reduce overstock and markdown waste. Better size recommendations can reduce the returns and unsold stock caused by poor fit.

- **SDG 9 – Industry, innovation and infrastructure.** The project shows how cloud-based AI services can be adopted by local retailers without dedicated beacon or camera hardware.
- **SDG 8 – Decent work and economic growth.** More efficient operations and better-targeted promotions support the competitiveness of a sector that employs many people in Sri Lanka, and staff-assistance alerts let employees spend their time where customers need them.

These are intended contributions. The project has not measured environmental or employment outcomes, and Section 7.5 discusses what would be required to evidence them.

1.4 Purpose and scope of the common report

This common report presents the shared problem, the integrated solution and the evidence that the four components operate together as one system. It covers the overall problem and its decomposition, the solution architecture, integration and shared data contracts, the common implementation, group-level evaluation, commercialization and responsible innovation. The detailed literature review, methodology, experiments and results of each component are presented in the four individual theses; this report summarizes them only as far as needed to explain the integrated system.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Multi-store fashion retailers make inventory, promotion, fitting and in-store service decisions using isolated tools and manual judgement, so decisions in one area are not informed by data from the others. As a result, stock is misallocated between stores, promotions are poorly targeted and may push products that are unavailable, customers face uncertainty about fit, and staff cannot identify shoppers who need help. The common problem addressed by this project is therefore the **absence of an integrated, predictive and explainable decision-support platform that connects stock, marketing, fitting and in-store behavior data for multi-store fashion retail**.

2.2 Root causes and overall research gap

Four root causes underlie the problem:

- **Reactive inventory management.** Stock levels and transfers are often planned manually from historical sales, without forecasts that account for promotions, seasonality, weather or store differences [6], [7].
- **Broad, stock-unaware marketing.** Campaigns are targeted using broad segments rather than individual redemption likelihood, and campaign tools are not aware of stock levels [9], [14].
- **Fit uncertainty.** Sizes vary between brands, and existing virtual try-on solutions typically separate measurement estimation from garment visualization [10], [15].
- **Invisible in-store behavior.** Indoor customer tracking usually requires costly beacon or camera infrastructure and manual monitoring [12].

Existing research and commercial solutions tend to address these causes individually. Forecasting, promotion targeting, customer tracking and virtual fitting are treated as separate systems, with limited integration, limited use of real-time behavioral input, and limited explainability of model decisions [16], [17], [11]. The overall research gap is a **unified platform in which predictive models for all four decision areas share data and identifiers, exchange outputs, and explain their recommendations to non-technical users**. No single component can close this gap, which is why the solution requires four interdependent components.

2.3 Subproblem and component mapping

The common problem was decomposed into four non-overlapping subproblems, each owned by one team member (Table 2.1). The subproblems are distinct in their data, models and users, but each depends on outputs from at least one other component.

Table 2.1: Subproblems and component mapping

ID	Subproblem	Component	Depends on / provides to
SP1	Future demand per product and store is unknown, and stock transfers are planned manually without considering cost or risk	C1 – AI-Driven Smart Inventory and Stock-Flow Optimization	Provides stock availability and overstock flags to C2
SP2	Promotions are not targeted at the customers most likely to redeem them, and may promote unavailable products	C2 – AI-Driven Personalized Marketing Intelligence	Uses stock data from C1; uses customer and transaction data shared with C3 and C4
SP3	Customers cannot predict their size across brands or see how garments will look	C3 – AI-Based Virtual Try-On and Size Recommendation	Uses product and inventory data from C1; contributes customer profile data used by C2
SP4	Staff cannot see customer movement or identify shoppers who need help	C4 – GPS-Based Customer Behavior Tracking and Staff Assistance	Uses customer identities shared with C2 and C3; contributes behavior data

All four are necessary. Without SP1, the marketing and styling components would recommend products that may not be in stock. Without SP2, inventory optimization cannot be linked to demand created by campaigns. Without SP3, the largest source of uncertainty in the customer's purchase decision remains. Without SP4, the system would have no view of in-store behavior and staff would remain reactive.

2.4 Overall aim and group objectives

Aim. To design, develop and evaluate an integrated AI-driven Smart Fashion Retail System that improves inventory, marketing, fitting and in-store service decisions for a multi-store fashion retailer through shared data and explainable predictions.

The group objectives, the components that deliver them and the evidence used to evaluate them are listed in Table 2.2.

Table 2.2: Group objectives, components and evaluation evidence

ID	Objective	Components	Evaluation evidence
O1	Forecast product demand per store with $R^2 \geq 0.90$ and generate explainable, risk-scored stock-transfer recommendations	C1	MAE, RMSE, R^2 on held-out data; transfer workflow tests
O2	Predict promotion redemption with accuracy above the majority-class baseline and generate promotional assets for high-affinity segments	C2, C1	Accuracy, ROC-AUC, confusion matrix; poster generation tests
O3	Predict body measurements and brand-specific sizes, and visualize garments on the customer's photograph	C3	MAE and R^2 per measurement; size accuracy, precision, recall, F1
O4	Classify in-store movement and raise staff-assistance alerts without dedicated tracking hardware	C4	Accuracy, F1, ROC-AUC; alert rule tests
O5	Integrate all four components through a shared back end, database and identifiers into one working platform	All	Integration and end-to-end tests (Section 6.2)
O6	Provide natural-language explanations of AI outputs for non-technical users	C1, C2, C3	Functional tests; user feedback

2.5 Scope assumptions and constraints

In scope. A multi-store fashion retailer with eight branches, approximately 1,000 products across six brands, a customer mobile/web application, a staff dashboard and a management dashboard. The four components, a shared FastAPI back end, Firebase Authentication and a Firebase Firestore database.

Out of scope. Integration with a live Point of Sale (POS) or Enterprise Resource Planning system, payment processing, supplier ordering, physical logistics execution, and outdoor or multi-floor indoor positioning infrastructure.

Assumptions. Models were trained and evaluated on partially simulated or prepared retail datasets that reflect realistic sales, campaign, body-measurement and movement patterns. Customers using the tracking features have consented to location sharing and keep the store application active. Stock transfers are executed only after human approval.

Constraints. The project was delivered within an academic year by a team of four, using free or low-cost tiers of cloud services. Generative services (Gemini, Imagen and Vertex AI) are external dependencies with usage quotas, and GPS accuracy indoors is limited by the hardware in customers' phones.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The Smart Fashion Retail System is one platform that supports a retailer's main operational decisions from a common data foundation. Its central value proposition is that **every recommendation the system makes — what to stock, what to promote, which size to buy, and whom to help — is predictive, explainable and consistent with the rest of the business**.

End to end, sales and stock data feed the inventory component, which forecasts demand and recommends transfers. The marketing component combines customer history with the same product and stock data to decide which customers to target with which promotions, and generates promotional posters. A customer who responds can use the virtual try-on component to find the right size and see the garment on their own photograph, while the AI stylist suggests in-stock items. When the customer visits a store, the tracking component monitors their movement and alerts staff when they appear to need help. Each transaction and behavioral event returns to the shared database and improves the next round of predictions.

3.2 High level architecture

The system follows a three-layer architecture (Figure 3.1 and Figure 3.2). The **presentation layer** is a Flutter Web application with customer, retail staff and administrator views. The **application layer** is a set of FastAPI services that expose the four intelligent components, authentication through Firebase Authentication, shared business logic and model serving. The **data layer** is a single Firebase Firestore database holding products, inventory, customers, campaigns, transactions, behavior data and model predictions [18]. External AI services — Gemini for explanations and image generation [19], [20] and Google Cloud Vertex AI for virtual try-on [21] — are called from the application layer only, so that credentials are never exposed to the client.

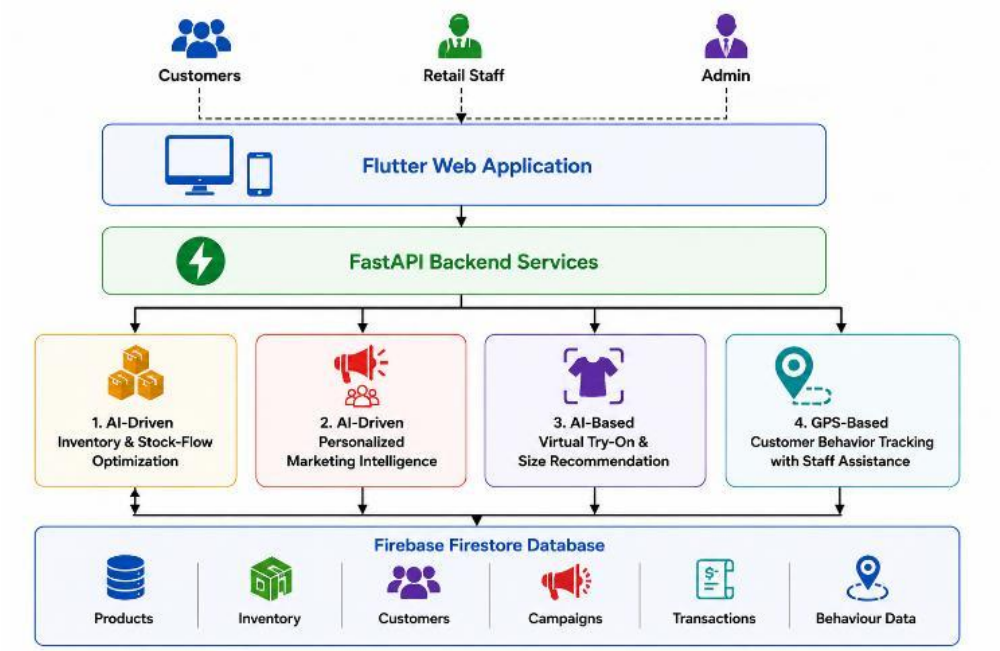


Figure 3.1: Overall architecture of the Smart Fashion Retail System

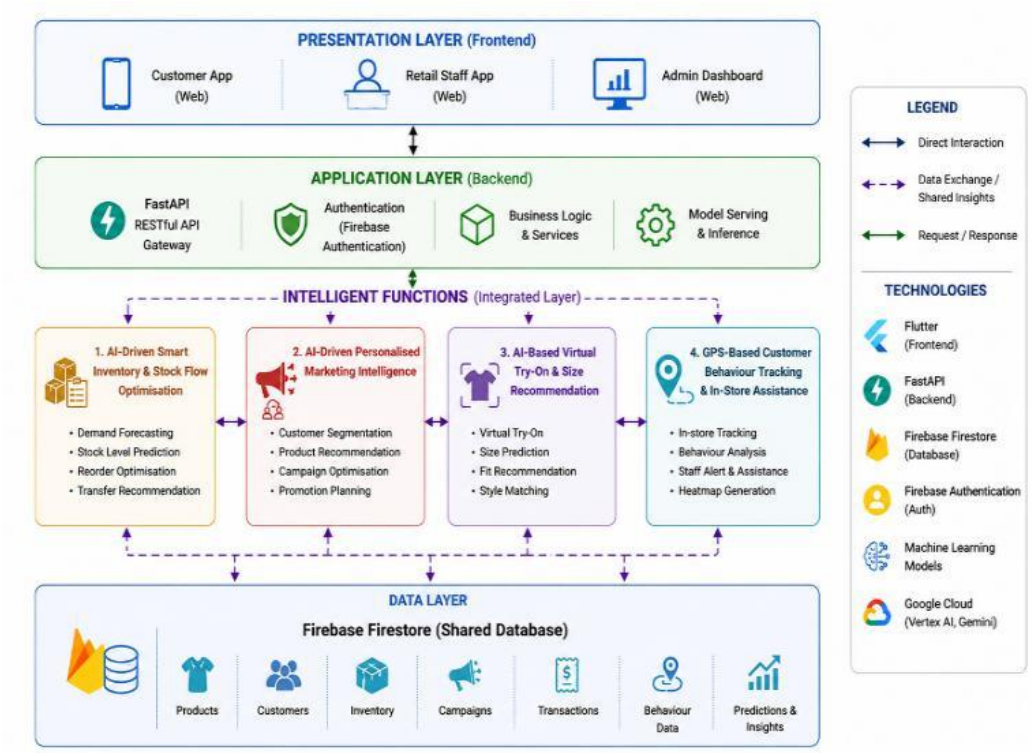


Figure 3.2: Layered integration view showing the presentation, application and data layers

3.3 Component 1 summary

AI-Driven Smart Inventory and Stock-Flow Optimization predicts future product demand and recommends stock movement between stores.

Item	Description
Owner	W M Methsara (IT22244598)
Subproblem	SP1 – misallocated stock and manual transfers
Inputs	Historical sales with lag-1 and lag-7 demand and seven-day rolling averages; product category, subcategory, brand, gender and price; promotion percentage; store; weather; calendar and festival variables
Core method	XGBoost regressor [5] for daily demand and monthly inventory intelligence; inventory-health rules (reorder status, days-on-hand); a ranking engine that scores source stores on available stock, predicted demand, safety stock, distance, logistics cost and shortage risk; a Gemini explainability layer
Outputs	Demand forecasts; inventory-health status; ranked transfer recommendations with confidence and risk scores; natural-language explanations; a review–approve–reject–cancel–execute workflow that updates inventory and movement history
Contribution to the system	Supplies stock availability and overstock information to the marketing component and in-stock product data to the AI stylist

3.4 Component 2 summary

AI-Driven Personalized Marketing Intelligence predicts which customers are likely to redeem which promotions and produces the campaign assets.

Item	Description
Owner	Wijetunga J.A.R.K. (IT22146588)
Subproblem	SP2 – broad, stock-unaware promotions
Inputs	Customer purchase and campaign-interaction history converted into RFM features [9]; product, promotion and customer attributes; stock availability from C1
Core method	Random Forest classifier [5] predicting promotion-redemption probability; segmentation and scheduling logic; Gemini Imagen 3 [20] for poster generation; Gemini 1.5 Flash [19] for explanations
Outputs	Redemption probabilities; target segments; campaign schedules aligned with salary cycles; promotional posters; explanations

Item	Description
Contribution to the system	Turns inventory signals into demand (for example, promoting overstocked items) and ensures that campaigns promote only products that are available

3.5 Component 3 summary

AI-Based Virtual Try-On and Size Recommendation acts as a digital fitting room.

Item	T A Silva
Owner	T A Silva (IT22284952)
Subproblem	SP3 – fit uncertainty across brands
Inputs	Height, weight and gender (with derived BMI); categorical product attributes; a customer photograph and garment image; product metadata, descriptions and reviews
Core method	Random Forest regressor [22] tuned with GridSearchCV (three-fold cross-validation) for shoulder width, waist circumference and leg length; 100-tree Random Forest classifier for brand-specific size; Vertex AI virtual try-on [21]; a RAG stylist [23] using all-MiniLM-L6-v2 embeddings [24], ChromaDB and LangChain
Outputs	Predicted measurements; recommended brand-specific size; fitted-garment image; product-grounded styling advice
Contribution to the system	Reduces purchase uncertainty for customers targeted by C2, recommends only in-stock items from C1, and enriches shared customer profiles

3.6 Component 4 summary

GPS-Based Customer Behavior Tracking and Staff Assistance monitors customer movement inside a store and prompts staff to help.

Item	Description
Owner	L A D A S Perera (IT22243812)
Subproblem	SP4 – invisible in-store behavior
Inputs	GPS coordinates from the customer application every three seconds; predefined store zones; manual SOS requests

Item	Description
Core method	Zone mapping; Haversine distance, velocity and direction-change features; 100-tree Random Forest classifier [22] for Browsing vs Transiting, with a rule-based speed fallback; dwell-time and pacing alert rules
Outputs	Customer intent class; staff-assistance alerts; customer locations on the staff dashboard; behavior records
Contribution to the system	Adds the in-store behavioral layer for the same customers served by C2 and C3, without beacon or camera infrastructure

3.7 End to end user and data flow

A typical journey through the integrated system is as follows:

- **Data ingestion.** Sales, stock, product, customer and campaign data are loaded into Firestore (including CSV upload and cleaning in the marketing dashboard) using common identifiers.
- **Inventory planning (C1).** The demand model forecasts sales per product and store. Where a shortage is predicted, the ranking engine proposes a transfer, which a manager reviews and approves; on execution, inventory and movement history are updated.
- **Campaign planning (C2).** The marketing component reads the updated stock and overstock data, scores customers' redemption probability, selects high-affinity segments, schedules the campaign and generates a promotional poster.
- **Customer engagement (C3).** A targeted customer opens the application, enters height and weight, receives a brand-specific size, uploads a photograph to see the garment, and asks the AI stylist for in-stock suggestions.
- **In-store visit (C4).** In the store, the application streams GPS positions. Prolonged browsing or pacing between zones triggers an alert on the staff dashboard, and the customer can also press SOS.
- **Feedback.** Purchases, redemptions and behavior records are written back to Firestore and become training and input data for the next forecasting and targeting cycle.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The team used a **shared-database, service-per-component** integration strategy. Each member owned the FastAPI router, models and Flutter screens for their component, while the Firestore schema, common identifiers, authentication and deployment configuration were agreed jointly and treated as shared assets. This allowed components to be developed in parallel while guaranteeing that they read and wrote the same entities.

Integration proceeded in stages: (1) agreement on the shared entities and identifiers; (2) independent development and testing of each component against the shared schema; (3) connection of the components to one Firestore project and one FastAPI application; (4) implementation of cross-component data exchanges, such as stock availability for marketing and in-stock retrieval for the stylist; and (5) end-to-end functional testing. The main integration risks — schema drift, conflicting identifiers and dependence on external AI services — were managed through the shared schema, code review on the common GitHub repository, and fallbacks described in Section 4.4.

4.2 Interfaces data contracts and shared models

Components communicate in two ways: synchronously through REST endpoints exposed by the FastAPI back end, which exchange JavaScript Object Notation (JSON) payloads, and asynchronously through shared Firestore collections. Table 4.1 lists the shared collections and their owners. Every document references entities through common identifiers — product, customer, store and transaction identifiers — so that, for example, a redemption recorded by the marketing component can be joined to the same customer's size profile and in-store behavior.

Table 4.1: Shared Firestore collections and data ownership

Collection	Key fields	Written by	Read by
products	product_id, brand, category, subcategory, gender, price	C1 (master data)	C1, C2, C3
inventory	product_id, store_id, stock level, reorder status	C1	C1, C2, C3

Collection	Key fields	Written by	Read by
customers	customer_id, profile, RFM features, measurements	C2, C3	C2, C3, C4
campaigns	campaign_id, product_id, offer, schedule, poster URL	C2	C2
transactions	transaction_id, customer_id, product_id, store_id, date	Data ingestion	C1, C2
behaviour data	customer_id, zone, timestamp, intent class, alerts	C4	C4, C2
predictions and insights	model, entity ID, prediction, explanation	C1, C2, C3, C4	Dashboards

The full list of endpoint groups, inputs and outputs is given in Appendix A. Validation rules include required identifiers on every write, numeric range checks on model inputs (for example, height and weight), and role checks through Firebase Authentication before any management action such as approving a transfer.

4.3 Shared nonfunctional requirements

Table 4.2: Shared non-functional requirements

Quality	Requirement and approach
Performance	Model inference is served in-process by FastAPI; GPS updates every three seconds must be classified quickly enough for timely alerts
Usability	One Flutter Web interface with role-specific views; natural-language explanations for non-technical users
Security	Firebase Authentication for all users; API keys for Gemini and Vertex AI held only on the server
Privacy	Location tracking only with customer consent; customer photographs used only for try-on; data minimization in line with the Sri Lankan Personal Data Protection Act (PDPA) [25]
Reliability	Rule-based fallback when the movement model is unavailable; human approval before stock transfers
Maintainability	Separate routers and services per component; shared schema; version control on GitHub
Scalability	Managed cloud database and stateless API services that can be scaled horizontally
Explainability	Gemini-generated explanations for inventory and marketing outputs; confidence and risk scores on recommendations [16], [17]

4.4 Integrated workflow and exception handling

In normal operation, the workflow described in Section 3.7 runs continuously: data written by one component becomes input for the next. The integrated design handles the following failure paths:

- **Movement model unavailable.** The tracking component switches to a rule-based speed classifier, so alerts continue at reduced accuracy.
- **Low-confidence or risky transfer.** Recommendations carry confidence and risk scores and cannot execute without manager approval; managers can reject or cancel them.
- **Product out of stock.** The marketing component and the AI stylist check stock data before recommending, so unavailable products are not promoted or suggested.
- **Changed customer data.** A change in stored height or weight automatically triggers recalculation of predicted measurements and size.
- **External AI service failure or quota limit.** The core machine-learning predictions (demand forecasts, redemption probabilities, measurement and size predictions, and movement classification) run locally within the FastAPI services and do not depend on external AI services, so they remain available when Gemini or Vertex AI is unreachable. In the marketing component, a failed or rate-limited Gemini image request returns an error to the user, who can retry once the service is available; event details and the brand logo are overlaid on posters locally using Pillow, so this step does not depend on the generative model. ⟦Inventory: confirm whether forecasts and transfer recommendations are still shown without the Gemini explanation.⟧ ⟦Virtual try-on: confirm what the customer sees if Vertex AI fails, e.g. the size recommendation without the fitted image.⟧ Handling these failures more gracefully, with user-friendly error messages and automatic retries, is identified as future work.
- **Customer needs urgent help.** The manual SOS option in the customer application bypasses the classifier and raises an alert immediately.

4.5 Team contribution and dependency matrix

Table 4.3 maps each team member to their component, their research and development responsibilities, and their shared contributions. Table 4.4 shows the data dependencies between components.

Table 4.3: Team contribution matrix

Member	Component and research	Development	Shared and integration work
W M Methsara	C1: demand forecasting and stock-transfer optimization	XGBoost models, ranking engine, transfer workflow, inventory dashboard	Set up the shared team Firebase project and inventory data; product and inventory schema; integration with shared inventory data (stock and overstock); repository integration (jointly). Poster and common reports
Wijetunga J.A.R.K.	C2: promotion-redemption prediction and generative marketing	RFM pipeline, XGBoost classifier, scheduling, poster generation, marketing dashboard	Customer and campaign schema; repository integration (jointly); product and inventory schema; integration with shared inventory data (stock and overstock); group paper, poster and common reports
T A Silva	C3: body-measurement and size prediction, virtual try-on, AI stylist	Random Forest models, Vertex AI integration, RAG pipeline, sizing screens	Customer mobile application; Firebase Authentication for the whole system; repository integration (jointly). Mobile application and the zonal spatial service integration. Poster and common reports

Member	Component and research	Development	Shared and integration work
L A D A S Perera	C4: movement classification and staff alerts	Zone-spatial service, trajectory classifier, alerting engine, staff dashboard	Behavior data schema; repository integration (jointly).). Mobile application and the zonal spatial service integration. Poster and common reports

Table 4.4: Component dependency matrix (row component uses data from column component)

Uses →	C1 Inventory	C2 Marketing	C3 Try-On	C4 Tracking
C1 Inventory	–	Campaign plans	–	–
C2 Marketing	Stock and overstock	–	Customer profiles	Behavior data
C3 Try-On	In-stock products	–	–	–
C4 Tracking	–	–	Customer identity	–

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The team selected a common stack (Table 5.1) so that all components could be deployed as one application. Flutter Web was chosen to provide customer, staff and management interfaces from a single code base. FastAPI was chosen because all four components are implemented in Python and needed to serve scikit-learn and XGBoost models directly. Firebase Firestore provides a managed, synchronized NoSQL database that requires no server administration [18], and Google's AI services provide generative capabilities that the team could not train itself [19], [20], [21].

Table 5.1: Common technology stack

Layer	Technology
Front end	Flutter Web (Dart)
Back end	FastAPI (Python)
Database	Firebase Firestore
Authentication	Firebase Authentication
Machine learning	XGBoost; Random Forest (scikit-learn); GridSearchCV
AI services	Gemini (explanations and image generation); Google Cloud Vertex AI (virtual-try-on-001); LangChain with ChromaDB and HuggingFace all-MiniLM-L6-v2 (RAG stylist)
Data processing	Pandas, NumPy
Version control and hosting	GitHub; GitHub Pages (image hosting)

5.2 Development and integration process

The project followed an iterative research and development methodology (Figure 5.1) comprising requirement analysis, data collection, preprocessing, model development, system integration and evaluation, with feedback from evaluation returning to earlier stages. Within this cycle, work was organized into two-week sprints planned in Microsoft Planner. The shared schema and identifiers were agreed before component development began, so that each member could build against a stable interface. Code

was managed in a shared GitHub repository (R26-IT-060) with one branch per member; changes were merged into the integration branch after peer review.

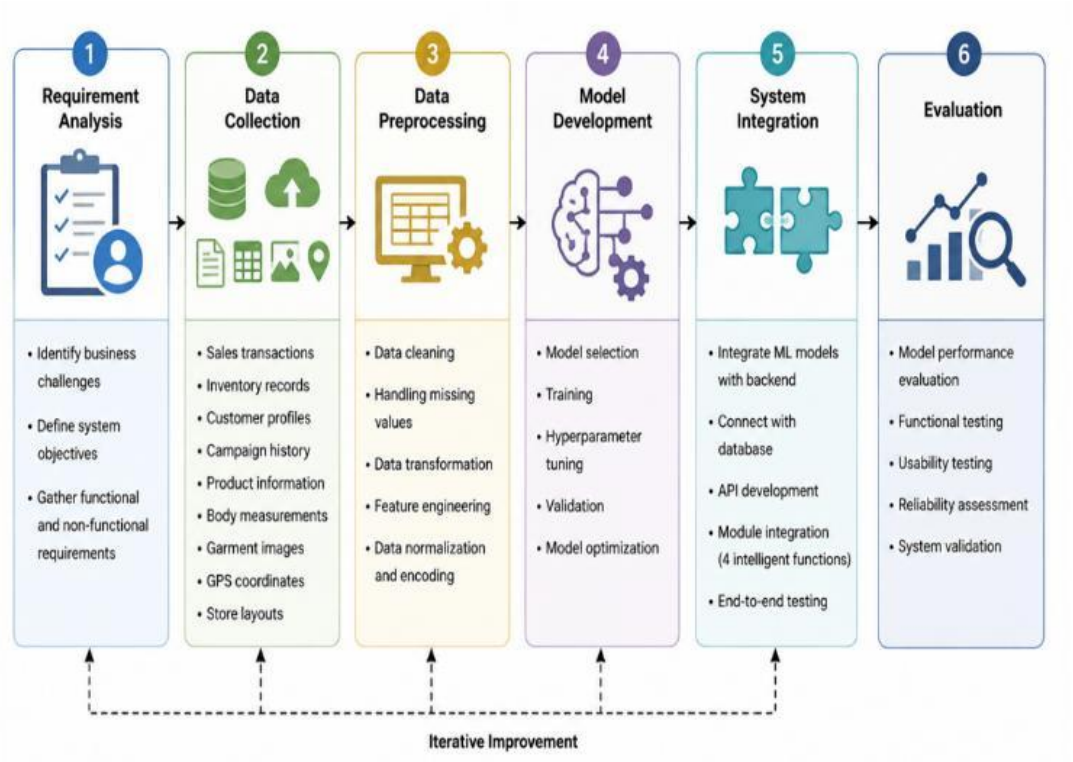


Figure 5.1: Research and development methodology of the Smart Fashion Retail System

5.3 Deployment and operational configuration

The integrated system is built as one FastAPI application with a router per component and one Flutter Web build. Configuration such as Firebase credentials and Google AI API keys is supplied through server-side environment variables. Trained model files are loaded at start-up. Firestore provides managed storage, and its export feature can be used for backup and restore. For the demonstration, the FastAPI back end and the Flutter Web front ends are run locally on the development machines, while data is stored in Firebase Firestore and the generative AI services (Gemini and Vertex AI) are called from the back end through Google's APIs. The front end of the virtual try-on component is additionally deployed on Azure Static Web Apps. Monitoring for the demonstration relies on the back-end application logs, the Firebase console for database usage, and the Google AI Studio and Google Cloud consoles for API usage

and quota limits. Full cloud deployment of the back end (for example, on Azure App Service or Google Cloud Run) with automated monitoring is identified as future work.

5.4 Evidence of team execution

Team execution is evidenced by the following records, summarized here and available on request:

- **Microsoft Planner**
- **Microsoft Teams**
- **GitHub (R26-IT-060)**.
- **Research diaries**

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The integrated system was evaluated against three group-level questions: (1) does each component's model meet its predictive success criterion; (2) do the components exchange data correctly and operate as one system; and (3) does the system meet the needs of its stakeholders? Component models were evaluated on held-out test data using the metrics in Table 6.1. Integration was evaluated through functional and end-to-end testing of the shared workflows.

Table 6.1: Evaluation metrics and success criteria

Component	Data	Metrics	Success criterion
C1	~297,011 records; 1,000 products; eight stores	MAE, RMSE, R^2	$R^2 \geq 0.90$
C2	Customer and campaign history (test set n = 6,000)	Accuracy, ROC-AUC, confusion matrix	Accuracy above majority-class baseline
C3	Body-measurement and historical fit data	MAE, R^2; accuracy, weighted precision, recall, F1	MAE ≤ 2 cm; accuracy ≥ 85%
C4	Movement trajectories (test set n = 250)	Accuracy, weighted precision, recall, F1, ROC-AUC	Accuracy ≥ 85%
System	Integrated workflows	Functional and end-to-end test pass rate	All critical workflows pass

6.2 Integration and end to end testing

The integrated system was validated through functional testing and performance evaluation to ensure reliability and interoperability. Table 6.2 lists the cross-component scenarios tested.

Table 6.2: Integration and end-to-end test scenarios

ID	Scenario	Components	Result
IT1	Approved stock transfer updates inventory, transactions and movement history	C1	Pass
IT2	Marketing excludes a product that the inventory component reports as out of stock	C1, C2	Pass

ID	Scenario	Components	Result
IT3	Overstock flag from inventory appears as a promotion candidate in marketing	C1, C2	Pass
IT4	AI stylist returns only in-stock products	C1, C3	Pass
IT5	Updated height/weight triggers measurement and size recalculation	C3	Pass
IT6	Browsing ≥ 30 s or A→B→A pacing within 60 s raises a staff alert	C4	Pass
IT7	Same customer ID is resolved consistently across C2, C3 and C4	C2, C3, C4	Pass
IT8	Unauthenticated user cannot approve transfers or view staff dashboard	All	Pass

6.3 Overall results

Table 6.3 summarizes the principal model results; detailed experiments are reported in the individual theses.

Table 6.3: Summary of component model results

Component	Model	Key results
C1	XGBoost – daily demand	MAE 0.5418; RMSE 0.9436; R^2 0.9497
C1	XGBoost – monthly inventory intelligence	MAE 14.213; RMSE 26.306; R^2 0.9353
C2	XGBoost – redemption classifier	Accuracy 85.5%; ROC-AUC 0.934
C3	Random Forest – body measurements	MAE 1.12 / 1.45 / 1.28 cm; R^2 0.914 / 0.897 / 0.921 (shoulder / waist / leg)
C3	Random Forest – brand-specific size	Accuracy 88.4%; weighted precision 0.879, recall 0.884, F1 0.881
C4	Random Forest – trajectory intent	Accuracy 91.2%; weighted precision 0.908, recall 0.912, F1 0.910; ROC-AUC 0.895

Inventory. The daily model explained about 95% of the variance in demand across eight branches. Recent and historical demand were the dominant predictors, with promotional, seasonal, customer and weather variables adding information (Figure 6.1). The forecasts feed shortage detection and transfer recommendations.

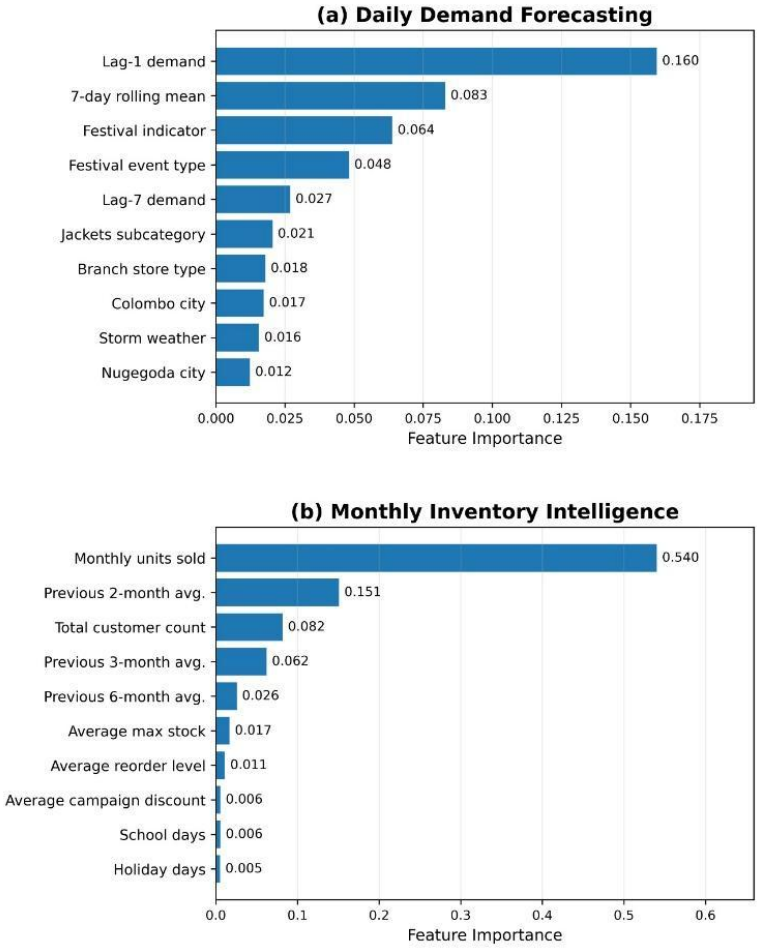


Figure 6.1: Feature importance of the inventory models: (a) daily demand forecasting and (b) monthly inventory intelligence

Marketing. On a test set of 6,000 customer–offer pairs the classifier correctly identified 3,042 redemptions and 2,090 non-redemptions (Figure 6.2), compared with a majority-class baseline of 55.9%. Switching the target from offer-type classification to redemption probability was the largest single contributor to performance. The ROC curve is shown in Figure 6.3.

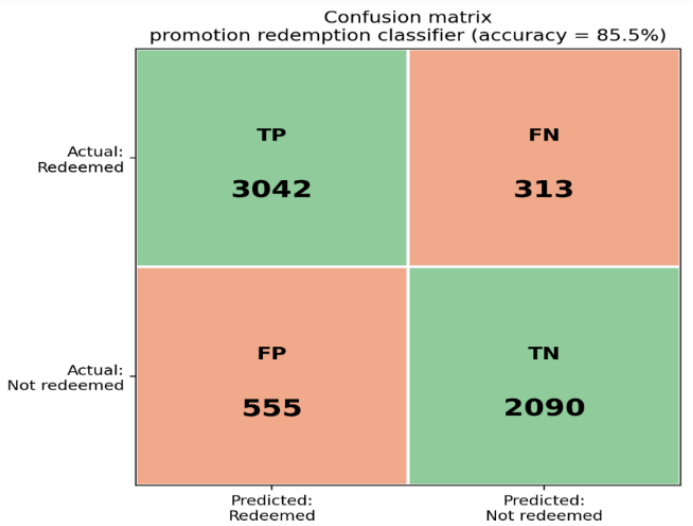


Figure 6.2: Confusion matrix of the promotion-redemption classifier

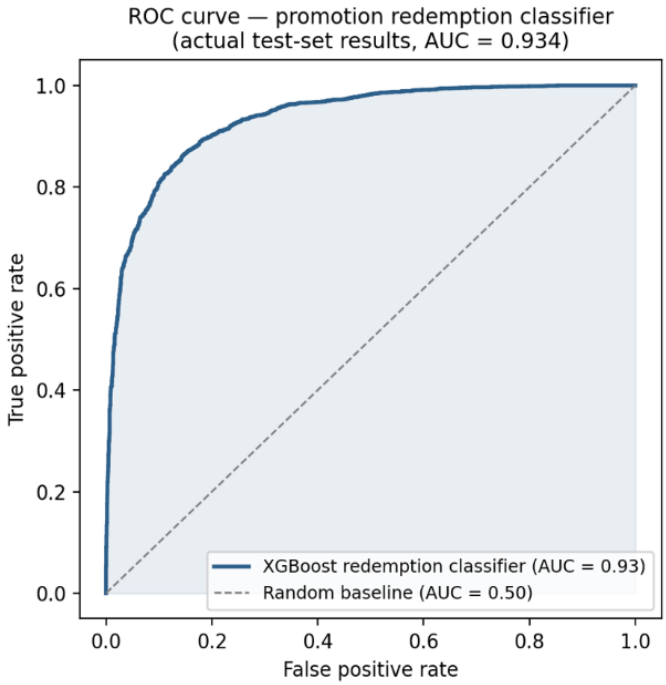


Figure 6.3: ROC curve of the promotion-redemption classifier

Virtual try-on. Body measurements were predicted to within about 1.5 cm on average (Figure 6.4), and brand-specific size was correct in 88.4% of cases.

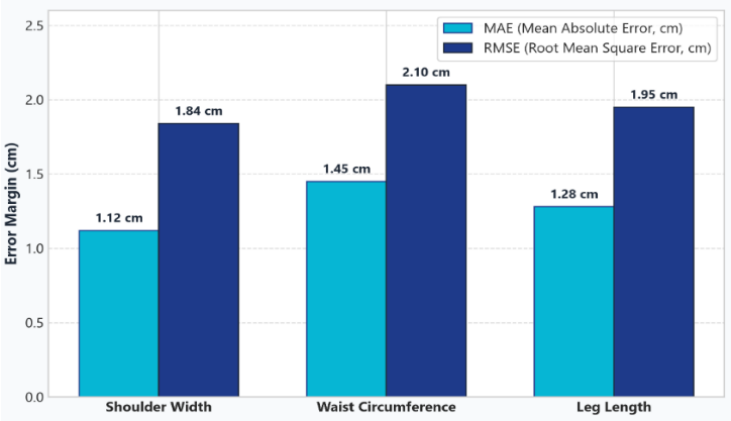


Figure 6.4: Body-measurement regression error by measurement

Behavior tracking. The trajectory classifier correctly classified 228 of 250 test trajectories as browsing or transiting (Figure 6.5), with balanced errors across the two classes.

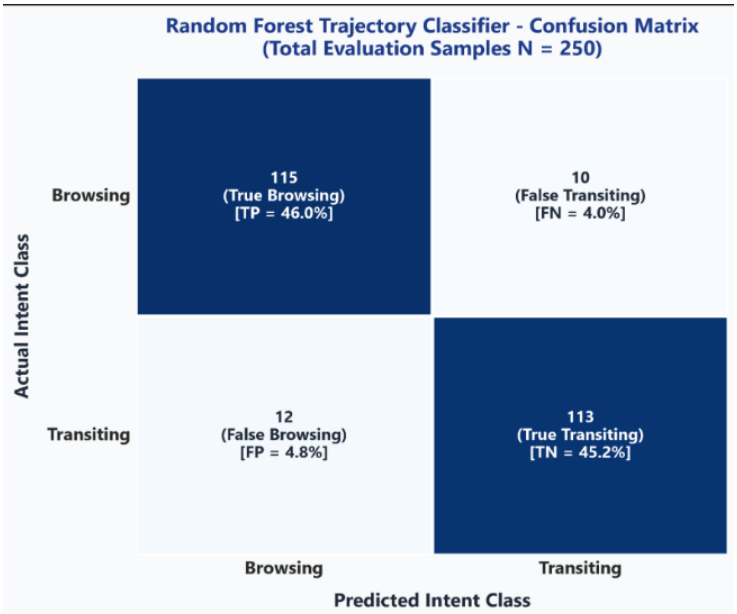


Figure 6.5: Confusion matrix of the Random Forest trajectory classifier

At system level, the results show that all four models exceeded their success criteria and that their outputs are available to the other components through the shared database.

6.4 Stakeholder and user validation

Formal stakeholder validation of the complete integrated system has not yet been carried out. Feedback to date has been informal and has come from three sources.

First, during requirement analysis, team members held informal conversations with retail marketing staff and small retail business owners about how promotions are planned, how stock is managed and how customers are assisted in store. These were exploratory discussions rather than a formal study, so no personal data was collected and no findings from them are reported as research results; they informed the choice of features and dataset attributes.

Second, the system was reviewed by the supervisors and the evaluation panel at the project's progress presentations. Third, classmates and peers tried the applications informally and commented on their ease of use.

This feedback led to two main changes. Natural-language explanations generated by Gemini were added so that non-technical users can understand model outputs such as forecasts and redemption predictions, and the user interfaces of the dashboards were revised to improve clarity and navigation. ⟦Optionally add one specific example of a UI change.⟧

Because none of this feedback came from a structured evaluation, it cannot be treated as evidence that the system meets stakeholder needs. A validation session is planned before final submission, in which ⟦number and type of participants, e.g. retail managers, store staff and customers⟧ will use the integrated system in a guided demonstration and complete ⟦a questionnaire, e.g. the System Usability Scale, and short interviews⟧ covering usefulness, ease of use and trust in the AI recommendations. The results and any changes made in response will be reported in the final version of this report.

6.5 Achievement of common objectives

Table 6.4: Achievement of group objectives

ID	Status	Justification
O1	Achieved	Daily R^2 0.9497 and monthly R^2 0.9353; transfer recommendations with confidence, risk and explanations implemented
O2	Achieved	Accuracy 85.5% against a 55.9% baseline; poster generation implemented
O3	Achieved	Measurement MAE 1.12–1.45 cm; size accuracy 88.4%; Vertex AI try-on implemented
O4	Achieved	Accuracy 91.2% without beacon or camera hardware; alert rules and SOS implemented

ID	Status	Justification
O5	Achieved	Shared FastAPI back end, Firestore database and identifiers; end-to-end tests in Table 6.2
O6	Partly achieved	Explanations implemented for inventory and marketing outputs; user understanding not yet formally evaluated

6.6 Integrated limitations and threats

The integrated system inherits the limitations of its components and adds some of its own:

- **Data validity.** All models were trained and evaluated on partially simulated or prepared data. Performance on live retail data may differ, and this is the main threat to external validity.
- **Coupling through shared data.** Because components rely on the same collections, errors in one component's data (for example, incorrect stock levels) propagate to the others.
- **External AI dependencies.** Explanations, posters and try-on images depend on Google services, their quotas and their pricing, and generated content is limited by the information supplied.
- **Indoor GPS accuracy.** GPS accuracy degrades indoors, continuous tracking affects battery life, and the application must remain active in the background.
- **Simplified operations.** Transfer costs use simplified logistics assumptions, and transfers still require human approval.
- **Generalizability.** The system was designed for one retailer configuration; other store layouts, brands or customer groups may require retraining.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

For a multi-store fashion retailer, the system offers one platform that reduces stock imbalance, increases the efficiency of promotional spending, reduces fit-related uncertainty and improves in-store service. Retail and inventory managers gain explainable forecasts and transfer recommendations; marketing teams gain predictive targeting and automated creative assets; customers gain accurate sizing, virtual try-on, relevant offers and faster help; and store staff gain visibility of where they are needed. Society benefits indirectly through reduced overstock and waste. This value proposition is informed by informal conversations with retail marketing staff and small retail business owners during requirement analysis, which highlighted manual stock transfers, broadly targeted promotions and limited visibility of in-store customer needs as recurring difficulties. These discussions were exploratory rather than a formal study, so the claimed benefits have not yet been measured; they will be assessed in the planned validation session described in Section 6.4.

7.2 Market alternatives and differentiation

The target segment is mid-sized multi-store fashion retailers in Sri Lanka and the region that lack the budget for enterprise retail suites. Alternatives include enterprise retail planning and customer data platforms, standalone e-mail and campaign tools, online size-recommendation plug-ins, and beacon- or camera-based in-store analytics. Each addresses only one of the four decision areas, often at a cost or hardware requirement unsuitable for local retailers. The system's differentiation is **integration** — one data model across inventory, marketing, fitting and in-store behavior — combined with **explainability** for non-technical users and **hardware-free** tracking using customers' own phones.

7.3 Business model and feasibility

The system could be offered as a subscription-based software service, priced per store, with optional modules so that a retailer can start with one component and add others. Key resources are the trained models, the shared platform and cloud infrastructure; key

partners are cloud and AI service providers and retail system integrators. The main costs are cloud hosting, database usage and generative AI calls, which scale with the number of stores and campaigns. Technical feasibility has been demonstrated by the working integrated prototype; operational feasibility depends on access to the retailer's sales, stock and customer data. Operating costs at the scale of a single retailer are modest and are driven mainly by the generative AI services. Firestore's no-cost quota (1 GiB of storage, 50,000 reads and 20,000 writes per day) covers the prototype's data volumes [26], and the Flutter front end can be hosted on the Azure Static Web Apps Free plan [27]. Virtual try-on is the largest variable cost at USD 0.12 per generated image [28], so 1,000 try-ons a month would cost about USD 120. Poster generation with Google's Nano Banana 2 Lite model costs about USD 0.034 per image, or roughly USD 10 for 300 posters, and natural-language explanations using Gemini 3.1 Flash-Lite cost about USD 3 for 3,000 explanations [29]. An indicative total is therefore around USD 135 per month, most of which scales with customer use of virtual try-on. Hosting the FastAPI back end in the cloud would add a further cost that depends on the provider and instance size. These figures are based on the providers' published prices in October 2026.

7.4 Adoption intellectual property and compliance

Adoption barriers include integration with existing POS and inventory systems, staff training, trust in AI recommendations and customers' willingness to share location data and photographs. Explanations, confidence scores and human approval steps are designed to build trust.

Intellectual property and licensing. The source code was developed by the team at SLIIT: hence the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work is granted .Open-source libraries (FastAPI, Flutter, scikit-learn, XGBoost, LangChain, ChromaDB) are used under their permissive licenses, and Google services are used under their terms of service.

Data protection and ethics. Customer location, body measurements and photographs are personal data. Deployment would require consent, purpose limitation, data

minimization, secure storage and retention limits consistent with the Sri Lankan PDPA [25] and location-privacy guidance [12]. Generated images and explanations should be labelled as AI-generated, and model decisions affecting customers should be open to review.

7.5 Sustainable Development Goal contribution

The most plausible contribution is to **SDG 12** through reduced overstock, better stock distribution and fewer fit-related returns, and to **SDG 8 and SDG 9** through more productive, digitally enabled retail operations [13]. Possible negative effects include privacy risks from tracking, increased consumption driven by more effective promotion, and staff concerns about monitoring. Sustaining the benefits would require measuring waste, returns and transfer volumes before and after deployment, keeping tracking opt-in, and using promotion to clear surplus stock rather than only to increase volume.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This project addressed the absence of an integrated, predictive and explainable decision-support platform for multi-store fashion retail. The team developed a Smart Fashion Retail System in which four components — inventory optimization, personalized marketing, virtual try-on with size recommendation, and GPS-based behavior tracking — share one back end, one database and common identifiers. Each component's model met its success criterion, and the components exchange data so that inventory informs marketing and styling, and customer, campaign and behavior data flow back for future predictions. The evidence shows that an integrated platform of this kind is technically feasible with a modest, cloud-based stack. Its value in live operations has yet to be demonstrated on real retail data.

8.2 Principal integrated contribution

Beyond the four components, the team's principal contribution is a **shared data model and service architecture that links stock, marketing, fitting and in-store behavior decisions**, together with cross-component behaviors that no single component provides: stock-aware promotion, in-stock styling advice, and a feedback loop from customer behavior to forecasting and targeting. The project also shows how generative AI can be used in a supporting role — explaining predictions and producing assets — around conventional, measurable ML models.

8.3 Future directions

- Pilot the integrated system with a partner retailer using live POS, stock and customer data, and re-evaluate all models and the integration under real conditions.
- Improve forecasting with hybrid models and additional data sources such as live traffic and POS data.
- Extend virtual try-on with larger training datasets and fine-tuned generative models.
- Evaluate alternative indoor-positioning technologies alongside GPS.
- Introduce continuous learning and model monitoring across all four components.
- Formally evaluate usability, trust and the effect of explanations with stakeholders.

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APPENDICES

Appendix A Integrated interface specification

Table A.1 summarizes the endpoint groups exposed by the integrated FastAPI back end.

Table A.1: Endpoint groups of the integrated back end

Component	Endpoint group	Input	Output
C1	Forecasting	product_id, store_id, date range	Predicted demand
C1	Decision generation and recommendation management	product_id, store_id	Ranked transfers with confidence and risk; approve, reject, cancel, execute actions
C1	Reference data and explanations	product_id, store_id	Product and store details; natural-language explanation
C2	Data upload and cleaning	CSV file	Cleaned records stored in Firestore
C2	Customer intelligence and redemption prediction	customer_id, promotion	Redemption probability; segment
C2	Scheduling and poster generation	Segment, product, offer	Campaign schedule; poster image
C3	Measurement and size prediction	Height, weight, gender, product	Measurements; brand-specific size
C3	Virtual try-on	Customer photo, garment image	Fitted-garment image
C3	AI stylist	Natural-language query	In-stock product suggestions
C4	Monitoring and alerts	customer_id, GPS coordinates, timestamp	Zone, intent class, alerts; staff dashboard data

Appendix B Team contribution declaration

Each member was responsible for the research, development and testing of their own component, as summarized in Table 4.3. All members contributed to the shared schema, integration, testing and the preparation of this report.

Table B.1: Team contribution declaration

Member	Student ID	Component	Signature
W M Methsara	IT22244598	AI-Driven Smart Inventory and Stock-Flow Optimization	
Wijetunga J.A.R.K.	IT22146588	AI-Driven Personalized Marketing Intelligence	
T A Silva	IT22284952	AI-Based Virtual Try-On and Size Recommendation	
L A D A S Perera	IT22243812	GPS-Based Customer Behavior Tracking and Staff Assistance	